

cyclepoint – mock setup

Mateja Perovic

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Required packages

```
library(gamm4)
library(ggplot2)
```

Load data

```
dat <- read.csv("yourData.csv")
```

Before modeling

Note that cyclepoint should always be a value between 0 and 1

A cyclepoint value higher than 1 indicates an issue in cycle day and/or length reporting. Always double-check your data.

Run this line of code once prior to model specification

```
cyc <- paste('cyclepoint [', toString(seq(0.0, 1.0, by=0.01)), ']' )
```

Mock examples

Cyclepoint only

bs='cc' sets the cyclic cubic spline basis function for cyclepoint.

random = ~ (1/subject) sets a random effect for participants (replace “subject” with your subject ID variable). Use this for repeated-measures designs or cross-sectional designs with nested measures (e.g., where each participant completes multiple conditions of a task). Remove this line if you have cross-sectional data without nested measures.

As shown in the example, covariates can be included in the model formula just as they would be in a linear model.

```

model <- gamm4(s(cyclepoint,bs='cc') + covariate1 + covariate2,
              random = ~(1|subject),
              data = dat)
summary(model$gam)

# Simple visualization:
plot(ggpredict(model$gam,terms=cyc))

```

With interaction term

Note that interaction terms are set up differently than in simple linear models. Add a *by =* command following the cyclepoint smooth setup, swapping out “interaction_term” with your predictor of interest. You may also include an additional *+ s(cyclepoint,bs='cc')* if you are interested in modeling main effects of cyclepoint in addition to the interactions of interest.

As shown in the example, covariates can be included in the model formula just as they would be in a linear model.

```

model <- gamm4(s(cyclepoint,bs='cc',by = interaction_term) + covariate1 + covariate2,
              random = ~(1|subject),
              data = dat)
summary(model$gam)

# Simple visualization:
plot(ggpredict(model$gam,terms=c(cyc,'interaction_term'))))

```

Interpreting output

Parametric coefficients: this section provides estimates for model terms that are treated in a similar way to the standard linear model (e.g., intercept, covariates). For example, if you control for depressive symptoms, this is where the effect will shown.

Approximate significance of smooth terms: this is the primary section of interest for cyclepoint analysis as it provides estimates for smooth/flexibly modeled terms. If you are only modeling effects of cyclepoint, you will see *s(cyclepoint)*. Similarly, if you are modeling interactions between cyclepoint and another predictor, this section will characterize the interaction. For example, if your interaction term is a two-level factor, you will see *s(cyclepoint):Level1* and *s(cyclepoint):Level2*.

EDF (estimated degrees of freedom; *edf* in model output) is an indicator of non-linearity. $\text{EDF} > 1$ suggests non-linearity, while $\text{EDF} \sim 1$ suggests a linear effect. Visualization of the smooth effect(s) is necessary for interpretation.

Simple visualization examples are provided above. We recommend checking out some of the validation examples provided in the toolkit (e.g., behavioral, physiological, neuroimaging) for more publication-ready plot examples.